

Clinical Data Quality & QC Analysis Transcript

***User Question:** Data quality/QC analysis in clinical data (ABP, ECG, and PPG waveform), things to look for to avoid typical clinical ML pitfalls? best python tools*

To successfully perform quality control (QC) and data quality analysis on physiological waveforms (ABP, ECG, and PPG), you must target artifacts unique to each modality and structural data flaws that sabotage machine learning models.

1. Modality-Specific Quality Control: What to Look For

Arterial Blood Pressure (ABP) Waveform

- **Sensor Damping:** Over-damped waveforms smooth out the dicrotic notch and underestimate systolic BP (SBP). Under-damped waveforms cause resonant 'ringing' artifacts, leading to overestimation of SBP.
- **Line Flushes & Blood Draws:** Catheter flushes cause sharp, rectangular high-pressure square waves followed by dead zones. These must be segmented and extracted out before training.
- **Hydrostatic Pressure Shifts:** Sudden baseline jumps occur if the physical pressure transducer is moved relative to the patient's heart level.

Electrocardiogram (ECG)

- **Baseline Wander:** Low-frequency drift caused by patient respiration or movement.
- **Powerline Interference:** 50/60 Hz electromagnetic noise from hospital grid equipment.
- **Electrode Motion Artifacts & Disconnection:** Sudden spikes or flatlines when leads lose contact with the skin, which can mimic ventricular fibrillation.

Photoplethysmogram (PPG)

- **Motion Artifacts (MA):** PPG is highly susceptible to patient movement. Micro-gap shifts between the optical sensor and the skin cause massive high-frequency distortion that masks the true heart rate.
- **Low Perfusion / Vasoconstriction:** If a patient is cold or on vasopressors, peripheral blood flow drops. This causes a severely degraded Signal-to-Noise Ratio (SNR), flattening out the crucial systolic peak and dicrotic notch.

2. Clinical ML Pitfalls to Avoid

- **The Cardinal Sin: Data Leakage via Subject Splitting:** In physiological datasets like MIMIC-III or VitalDB, multiple windows or recordings often belong to the same patient. If you use a random shuffle split, the model memorizes patient-specific baselines (e.g., a specific patient's unique pulse wave velocity or arterial stiffness) instead of learning generalized cardiovascular physics. **Always split your data strictly subject-wise (by patient ID).**

- **Signal Desynchronization:** If your project involves cross-modality estimation (e.g., using ECG + PPG to predict ABP), sub-millisecond clock drifts between sensors will corrupt metrics like Pulse Arrival Time (PAT). Ensure your datasets feature perfectly synchronized timestamps.
- **The 'Normal Distribution' Trap:** Clinical data heavily skews toward healthy baselines. If you are predicting hypotension or hypertension, a model can look deceptively accurate simply by predicting the average, normal MAP. Use stratified sampling and robust loss functions.

3. Best Python Tools for Waveform QC & ML Preprocessing

- **vital_sqi:** Out-of-the-box Signal Quality Indices (SQIs) for ECG and PPG. Computes over 70 distinct SQIs, contains built-in peak detectors, and creates rule-based thresholds to automatically reject noisy segments.
- **NeuroKit2:** Comprehensive physiological signal processing (ECG, PPG, EDA). Excellent for baseline wander removal, clean QRS/systolic peak detection, and extracting continuous heart rate variability (HRV) metrics.
- **pyPPG:** Advanced, specialized PPG feature extraction. Detects fiducial points (systolic/diastolic peaks, notches) and extracts derivative features (VPG/APG) critical for cuffless blood pressure models.
- **SciPy (Signal):** Fundamental signal architecture. Designing custom Butterworth bandpass filters (e.g., 0.5-40 Hz for ECG, 0.5-8 Hz for PPG) and notch filtering for powerline noise.

Summary Checklist for Your Pipeline

1. Apply bandpass filtering to remove baseline drift and high-frequency noise.
2. Segment data into fixed windows (e.g., 10 to 30 seconds).
3. Run a library like vital_sqi to calculate per-window quality scores and discard segments falling below your threshold.
4. Implement a strict GroupKFold cross-validation scheme based on Patient IDs to completely eliminate data leakage.

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User Question: *include my questions*

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